

①

Recall (F, R, k) system.

Thm (Brauer - Nesbitt) Let (F, R, k) be a p -module system, G be a finite group. U be a finite generated FG -module. ~~Let~~ L_1 and L_2 be full RG -lattices in U . Then $L_1/\pi L_1$ and $L_2/\pi L_2$ have the same composition factors with the same multiplicities as kG -modules.

proof: • observe: $L_1 + L_2$ is also a full RG -lattice in U

$\Rightarrow$ we can assume ~~$L_1 \subseteq L_2$~~ $L_1 \subseteq L_2$

• As R -modules, L_1 and L_2 are free and of the same rank

$\Rightarrow L_2/L_1$ is a torsion R -module

So L_2/L_1 has a composition series as a R -module and ~~hence~~ also as an RG -module. Because every series of RG -modules can be extended to give a composition series of R -modules.

$\Rightarrow$ We can assume that L_1 is a maximal submodule of L_2

Thus L_2/L_1 is simple $\Rightarrow \pi L_2 \subseteq L_1$

So $\pi L_1 \subseteq \pi L_2 \subseteq L_1 \subseteq L_2$. It suffices to show

$$L_2/L_1 \cong \pi L_2/\pi L_1$$

In fact: $L_2 \rightarrow \pi L_2/\pi L_1$ is a surjective morphism with
 $x \mapsto \pi x + \pi L_1$ ker = L_1

Let (F, R, k) splitting for G . The decomposition matrix D is the matrix with row indexed by the simple FG -modules and columns indexed by the simple kG -modules where entries are the numbers d_{TS} = multiplicity of S as a composition factor of $T_0/\pi T_0$ where S is a simple kG -module, T is an FG -module and T_0 is a full RG -lattice in T . d_{TS} is called the decomposition number.

Let (F, R, k) splitting for G , R complete

Define $G_0(FG)$ the free abelian group with the ~~isomorphism~~ isomorphism types of simple FG -modules as basis

Analogly we define $G_0(kG)$

$K_0(kG)$ indecomposable projective kG -modules as basis

If T is a simple FG -module, write $[T]$ the corresponding basis element of $G_0(FG)$. If S is

a simple kG -module, write $[S]$ for $G_0(kG)$

If P is a indecomposable projective kG -module

write $[P]$ for $K_0(kG)$

If U is a kG -module with composition ~~series~~ ②
factors $S_1, \dots, S_r$ with multiplicities

$n_1, \dots, n_r$, write $[U] = n_1[S_1] + \dots + n_r[S_r]$ in $G_0(kG)$

Similarly for an FG -module V if

$V \cong T_1^{n_1} \oplus \dots \oplus T_s^{n_s}$ write $[V] = n_1[T_1] + \dots + n_s[T_s]$ in $G_0(FG)$

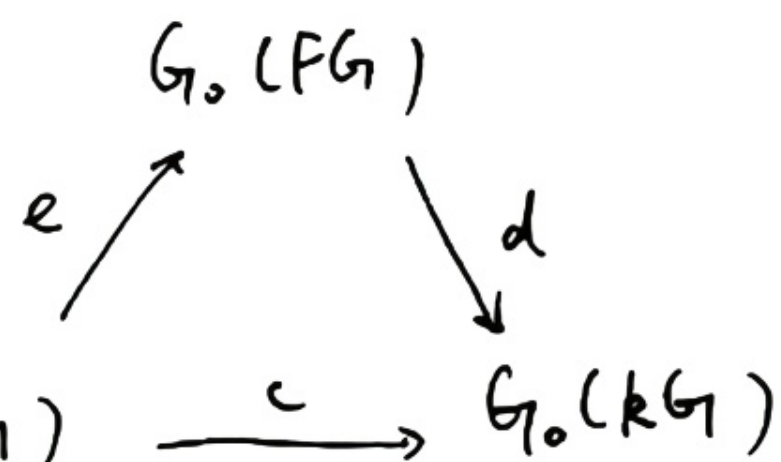
If P is a projective kG -module with

$P \cong P_1^{n_1} \oplus \dots \oplus P_s^{n_s}$ where $P_1, \dots, P_s$ are indecomposable projective kG -modules write $[P] = n_1[P_1] + \dots + n_s[P_s]$

in $K_0(kG)$

Consider cde triangle:

If P_s is a finitely generated indecomposable projective kG -module



then $e([P_s]) = [F \otimes_R \hat{P}_s] \in G_0(FG)$

where $\hat{P}_s$ is an RG -module that lifts P_s

recall: $\hat{P}_s$ is unique by isomorphism

• If V is a simple FG -module containing a full ~~sub~~

• ~~If V is~~ RG -lattice V_0 Put

$d([V]) = [V_0 / \pi V_0] \in G_0(kG) \Rightarrow$ the matrix of d is

the transpose of decomposition matrix

Let P_s be an indecomposable projective kG -module that is the projective cover of a simple kG -module $\hookrightarrow$

define $c([P_S]) = [P_S]$ The matrix of c = Cartan matrix

$$[P_i] = \sum_{\text{simple } S} C_{ST} [S] \quad \text{where } S, T \text{ simple } kG\text{-module}$$

Thm: $c = de$

proof: Let P_S be a indecomposable projective kG -module

$$c([P_S]) = [P_S] \quad e([P_S]) = [F \otimes_R P_S^1]$$

$$de([P_S]) = d([F \otimes_R P_S^1]) \quad \text{choose a full } R_G\text{-lattice}$$

$$\text{reducing by } \pi, \quad d([F \otimes_R P_S^1]) = [P_S]$$

proposition: (F, R, k) p -modular system. U, V F_G -modules containing full R_G -lattice U_0 and V_0

(i). $\text{Hom}_{R_G}(U_0, V_0)$ is a full R_G -lattice in $\text{Hom}_{F_G}(U, V)$

(ii). If U_0 is projective R_G -lattice, then

$$\text{Hom}_{R_G}(U_0, V_0) / \pi \text{Hom}_{R_G}(U_0, V_0) \cong \text{Hom}_{R_G}(U_0, V_0 / \pi V_0)$$

$$\cong \text{Hom}_{k_G}(U_0 / \pi U_0, V_0 / \pi V_0)$$

proof: i. First: $\text{Hom}_{R_G}(U_0, V_0)$ can be viewed as a subset of $\text{Hom}_{F_G}(U, V)$ take R -bases $u_1, \dots, u_r$ and $v_1, \dots, v_s$ for U_0, V_0 respectively that are also F -bases for U and V

Any R_G -morphism $U_0 \rightarrow V_0$ can be represented as a matrix over R . Regard it as a matrix over F

(3)

represents an F_G -morphism $U \rightarrow V$

To see $\text{Hom}_{R_G}(U_0, V_0)$ is a lattice in $\text{Hom}_{F_G}(U, V)$

$$\text{Hom}_{R_G}(U_0, V_0) \subseteq \text{Hom}_R(U_0, V_0) \simeq R^{rs}$$

The lattice is R -free $\Rightarrow \text{Hom}_{R_G}(U_0, V_0)$ is R -free ~~since~~
 R is PID

Next: show $\text{Hom}_{R_G}(U_0, V_0)$ is a full R_G -lattice of $\text{Hom}_{F_G}(U, V)$

Let $\phi: U \rightarrow V$ be an F_G -morphism, then $\phi(u_i) = \sum \lambda_{ji} u_j$, $\lambda_{ji} \in F$. Choose $a \in R$ s.t. $a \lambda_{ji} \in R$ for all j, i .

Then $a\phi: U_0 \rightarrow V_0$ is an R_G -morphism, $\phi \in F \cdot \text{Hom}_{R_G}(U_0, V_0)$
 $\Rightarrow \text{Hom}_{R_G}(U_0, V_0)$ spans $\text{Hom}_{F_G}(U, V)$ over F

14. U_0 is projective

$$\Rightarrow \text{Hom}_{R_G}(U_0, \pi V_0) = \pi \text{Hom}_{R_G}(U_0, V_0)$$

$$\text{Consider } \text{Hom}_{R_G}(U_0, V_0) \rightarrow \text{Hom}_{R_G}(U_0, V_0 / \pi V_0)$$

U_0 projective $\Rightarrow$ it is an epimorphism

$$\text{Its kernel} = \text{Hom}_{R_G}(U_0, \pi V_0) = \pi \text{Hom}_{R_G}(U_0, V_0)$$

$$\text{So } \text{Hom}_{R_G}(U_0, V_0) / \pi \text{Hom}_{R_G}(U_0, V_0) \simeq \text{Hom}_{R_G}(U_0, V_0 / \pi V_0)$$

For the second isomorphism, all morphism $U \rightarrow V_0 / \pi V_0$ containing πU_0 in its kernel

$$\text{and so factor through } U_0 \rightarrow U_0 / \pi U_0 \rightarrow V_0 / \pi V_0$$

corollary: Suppose U_0, V_0 are full R_G -lattices in U, V
 where U, V FG -modules

U_0 : projective

Then

$$\dim_F \operatorname{Hom}_{FG}(U, V) = \dim_k \operatorname{Hom}_{kG}(U_0/\pi U_0, V_0/\pi V_0)$$

proof: Both equal to the R -rank of $\operatorname{Hom}_k(U_0, V_0)$

Thm: (F, R, k) splitting for G . R : complete

(1) Let S be a simple kG -module. T : simple FG -module
 containing a full R_G -lattice T_0

The multiplicities of T in $F \hat{\otimes}_R P_S$ = the multiplicity
 of S as a composition factor in $T_0/\pi T_0$

(2) The matrix of e is D . the matrix of d is D^T
 with respect to basis where D is the decomposition
 matrix

proof: (1). $\dim_F \operatorname{Hom}_{FG}(F \hat{\otimes}_R P_S, T) = \dim_k \operatorname{Hom}_{kG}(P_S, T_0/\pi T_0)$

left side: $\dim_F \operatorname{End}_{FG}(T) \cdot (\text{multiplicity of } T \text{ in } F \hat{\otimes}_R P_S)$

right side: $\dim_k \operatorname{End}_{kG}(S) \cdot (\text{multiplicity of } S \text{ as a
 composition factor of } T_0/\pi T_0)$

splitting $\Rightarrow \dim_F \operatorname{End}_{FG}(T) = \dim_k \operatorname{End}_{kG}(S) = 1$

(2). obviously

Condition: (F, R, k) splitting for G , R : complete

(4)

Then the Cartan matrix $C = D^{-1}D$

In particular, C is a symmetric matrix

Let S be a commutative ring with 1. $R \subseteq S$ a subring of S . $1 \in R$

An element $s \in S$ is called integral over R if $f(s) = 0$ for some monic polynomial $f \in R[x]$. Recall following results.

Thm: (1). TFAE for $s \in S$

(a) s is integral over R

(b) $R[s]$ is contained in some R -submodule M of S which is finite generated over R s.t

$$sM \subseteq M$$

(2) The elements of S integral over R form a subring of S

$$(3) \{x \in \mathbb{Q} \mid x \text{ is integral over } \mathbb{Z}\} = \mathbb{Z}$$

Complex (4). Let $g \in G$, G finite group. Let χ be a character of G . Then $\chi(g)$ is an algebraic integer.

(Recall that: An element of $\mathbb{C}$ is called algebraic integer if it is integral over $\mathbb{Z}$)

Consider $\mathbb{C}G \supseteq \mathbb{Z}G$

$$\mathbb{Z} := \mathbb{Z} \cdot 1_G \subseteq \mathbb{Z}(\mathbb{Z}G)$$

proposition: The center $\mathbb{Z}(\mathbb{Z}G)$ is integral over $\mathbb{Z}$

Hence if $G_1, \dots, G_r$ be conjugacy classes of G

$$\bar{C}_i = \sum_{g \in G_i} g, \text{ let } \lambda_1, \dots, \lambda_r \text{ be algebraic integers in}$$

$\mathbb{C}$ Then $\sum_{i=1}^r \lambda_i \bar{C}_i$ is integral over $\mathbb{Z}$

proof: $\mathbb{Z}(\mathbb{Z}G) \simeq \bigoplus_{i=1}^r \mathbb{Z}\bar{C}_i$ ~~is a finite generated~~

finite generated abelian group $\Rightarrow$ integral over $\mathbb{Z}$

Let $\rho_1, \dots, \rho_r$ be simple $\mathbb{C}G$ -representation of G , of degree $d_1, \dots, d_r$ with characters $\chi_1, \dots, \chi_r$

$$\rho_i: \mathbb{C}G \rightarrow GL_{d_i}(\mathbb{C})$$

extend to $\rho_i: \mathbb{C}G = \bigoplus_{i=1}^r M_{d_i}(\mathbb{C}) \rightarrow M_{d_i}(\mathbb{C})$ is the projection.

proposition: Fix i , let $x \in \mathbb{Z}(\mathbb{C}G)$, then $\rho_i(x) = \lambda I$ for some $\lambda \in \mathbb{C}$

$$\text{In fact } \rho_i(x) = \frac{1}{d_i} \text{tr}(\rho_i(x)) I \text{ write } x = \sum_{g \in G} a_g g$$

$$\text{then } \rho_i(x) = \frac{1}{d_i} \sum_{g \in G} a_g \chi_i(g) \cdot I$$

proof: ~~$xy = yx$~~ $xy = yx, \forall y \in \mathbb{C}G \Rightarrow \rho_i(y)\rho_i(x) = \rho_i(x)\rho_i(y)$

for any $y \in \mathbb{C}G$ and for any i

by Schur's Lemma. $\rho_i(x)$ is a scalar matrix

$$\begin{aligned}
 \rho_i(x) &= \lambda I & \lambda &= \frac{1}{d_i} \text{tr}(\rho_i(x)) = \frac{1}{d_i} \text{tr}\left(\sum_{g \in G} a_g \rho_i(g)\right) \\
 & & &= \frac{1}{d_i} \sum_{g \in G} a_g \text{tr}(\rho_i(g)) \\
 & & &= \frac{1}{d_i} \sum_{g \in G} a_g \chi_i(g)
 \end{aligned}
 \tag{4}$$

Thm: The degrees d_i of simple complex representations of G divide $|G|$

proof: let $\chi = \sum_{g \in G} \overline{\chi_i(g)} \cdot g$ then $\chi \in \mathbb{Z}(\mathbb{C}G)$

$$\begin{aligned}
 \rho_i(x) &= \frac{1}{d_i} \left(\sum_{g \in G} \overline{\chi_i(g)} \chi_i(g) \right) \cdot I \\
 &= \frac{|G|}{(d_i)} I
 \end{aligned}$$

$\overline{\chi_i(g)} = \chi_i(g^{-1})$ are ~~integral~~ algebraic integers, $\forall g$

$\chi = \sum_{c \in \mathbb{C}G} \overline{\chi_i(g_c)} \left(\sum_{g \in G} g \right) \in \mathbb{Z}(\mathbb{C}G)$ is integral

over $\mathbb{Z}$

$\rho_i(x)$ is integral over $\mathbb{Z}$

$\frac{|G|}{d_i} \cdot 1$ is integral over $\mathbb{Z}$, but $\frac{|G|}{d_i} \in \mathbb{Q}$

hence $\frac{|G|}{d_i} \in \mathbb{Z}$.

Sample S_3 character table

- linear character of $G = \text{irr character of } G / [G, G]$
- $|G| = \sum_{\chi \in \text{Irr}(G)} \chi(1)^2$
- $\chi(1) \mid |G|, \forall \chi \in \text{Irr}(G)$
- $|\text{U}(G)| = |\text{Irr}(G)|$
- orthogonal ~~con~~ relation

linear character of $G = S_3$.

$$S_3 / [S_3, S_3] = S_3 / A_3 = C_2$$

	(1)	(12)	(123)
1	1	1	1
ϵ	1	-1	1
χ	2	0	-1

Sample. D_8

	(1)	(2)	(3)	(4)	(5)
χ_1	1	1	1	1	
χ_2	1	-1	1	-1	
χ_3	1	1	-1	-1	
χ_4	1	-1	-1	1	
χ_5	2				